CS 171: Discussion Section 7 (October 20)

1 CPA-Secure Public-Key Encryption From Two-Round Key Exchange

Question: Given a two-round key-exchange protocol with keyspace $\mathcal{K} = \{0,1\}^n$, construct a CPA-secure public-key encryption (PKE) scheme for n-bit messages and prove its security. Do not use any other cryptographic primitive.

1.1 Two-Round Key Exchange

A two-round key-exchange protocol comprises three randomized algorithms (P_1, P_2, P_3) and has the following form:

1. Alice computes $(\mathsf{msg}_1, \mathsf{st}) \leftarrow P_1(1^n)$ and sends msg_1 to Bob.
2. Bob computes $(\mathsf{msg}_2, k) \leftarrow P_2(\mathsf{msg}_1)$. Then he sends msg_2 to Alice and outputs k.
3. Alice computes $k \leftarrow P_3(\mathsf{st}, \mathsf{msg}_2)$ and outputs k.

1.1.1 Definition of Security

We will define security for key exchange below. Our definition of security is equivalent to the one given in lecture 14, slide 16.

Consider the following security game.

$\underline{G_{\mathcal{B},\Pi}(n,b)}$:

1. The challenger executes the key exchange protocol Π to produce $(\mathsf{msg}_1, \mathsf{msg}_2, k)$.
2. If $b = 0$, the challenger sets $\hat{k} = k$. If $b = 1$, they sample $\hat{k} \leftarrow \mathcal{K}$. Then the adversary $\mathcal{B}$ is given $(\mathsf{msg}_1, \mathsf{msg}_2, \hat{k})$.
3. $\mathcal{B}$ outputs a bit b', which is the output of the game as well.

We say that a key-exchange protocol is **secure** if for all PPT adversaries $\mathcal{B}$, there exists a negligible function negl such that:

$$\left| \Pr[G_{\mathcal{B},\Pi}(n,0) \to 1] - \Pr[G_{\mathcal{B},\Pi}(n,1) \to 1] \right| = \mathsf{negl}(n)$$

1.2 Definition of CPA security for PKE

Let's write the definition of CPA security for public-key encryption. It will resemble the definition we've seen previously for secret-key encryption.

Given an adversary $\mathcal{A}$, define the following game:

$\underline{\mathsf{PubK}_{\mathcal{A},\Pi}(n)}$:

1. The challenger samples the keys $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Gen}(1^n)$. Then they give $(1^n, \mathsf{pk})$ to the adversary $\mathcal{A}$.
2. $\mathcal{A}$ outputs a pair of messages (m_0, m_1) such that $|m_0| = |m_1|$.
3. The challenger samples $b \leftarrow \{0, 1\}$ and computes the challenge ciphertext:

$$c \leftarrow \mathsf{Enc}(\mathsf{pk}, m_b) \tag{1.1}$$

   Then they give c to $\mathcal{A}$.
4. $\mathcal{A}$ outputs a bit b'. The output of the experiment is 1 if $b = b'$ and 0 otherwise.

A public-key encryption scheme is **CPA-secure** if for any probabilistic polynomial-time adversary $\mathcal{A}$, there is a negligible function negl such that:

$$\Pr[\mathsf{PubK}_{\mathcal{A},\Pi}(n) \to 1] = \frac{1}{2} + \mathsf{negl}(n)$$

2 One-way functions from Pseudorandom Permutations

One-way functions can be constructed from many other cryptographic primitives, including from pseudorandom permutations.

Let $F : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}^n$ be a pseudorandom permutation. This can be written as $F(k,x)$ or equivalently $F_k(x)$, where k is the key. Note that an adversary can compute $F_k^{-1}(\cdot)$ in addition to $F_k(\cdot)$ if they are given the key k.

1. Let $x \in \{0,1\}^n$, and
$$\text{let } f_1(x) = F_{0^n}(x)$$
Show that f_1 is not a one-way function.

2. Let $x = (x_0, x_1) \in \{0,1\}^n \times \{0,1\}^n$, and
$$\text{let } f_2(x) = F_{x_0}(x_1)$$
Show that f_2 is not a one-way function.

3. Extra problem: Let $x \in \{0,1\}^n$, and
$$\text{let } f_3(x) = F_x(0^n)||F_x(1^n)$$
Show that f_3 is a one-way function.